

Daily operation

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NOTE

This chapter now draws on HydroVolta's own *HMI Description* specification (revision V5, Morocco project, dated 2026-06-16) — the design document the automation team is building the PLC program against. It describes screens, controls and workflow *as specified*; the PLC program itself was still in development at the time of writing (chapter 1), so exact on-screen wording/layout on a specific delivered unit may differ slightly from what is described here. Where an older pilot's tag names differ from this specification (e.g. TSA-prefixed tags used elsewhere in this manual), both are noted.

6.1 Operating elements

The local control panel exposes physical Start, Stop and Reset controls, a Manual/local selector, a feedwater-source selector, and a CIP-mode input (control-system tags `Start`, `Stop`, `Reset`, `Manual`, `FW_Select`, `LS_Select`, `CIP`), alongside a four-color status LED stack (`GREEN_LED`, `BLUE_LED`, `ORANGE_LED`, `RED_LED`). The HMI specification additionally confirms a **hardware Emergency Stop button that bypasses the PLC entirely** — see section 6.9.



Figure 6.1. Suggested figure: photograph of the local control panel with each button, selector switch and LED labeled and cross-referenced to the tag names above.

6.2 Plant modes

Per the HMI specification, the unit operates in one of nine defined modes, selected from the **Mode Control** screen (section 6.4):

1. **Initial fill** — automatic filling of the system before first production.
2. **Production (forward)** — normal automatic production.
3. *(reversal is a timed sub-state of production, not a separate numbered mode — see Starting the unit below)*
4. **Controlled shutdown** — an orderly stop sequence.
5. **CIP** — automatic clean-in-place, after a CIP-PREP step (High pH/NaOH or Low pH/HCl, chapter 12) has conditioned the loop.
6. **Flush/rinse** — freshwater flush of the process loop.
7. *(reserved)*
8. **Manual** — individual device control, access-gated (see *Manual mode* below).
9. *(reserved)*

A separate **Safe Stop** state exists outside the numbered mode list: it is entered automatically on certain faults and requires the recovery procedure in *Emergency stop and restart* below before the operator can select a new mode.

6.3 Starting the unit

Per the HMI specification, starting the unit is a two-step mode selection from the **Mode Control** screen: first **Start AUTO INITIAL FILL**, then, once initial fill completes, **Start AUTO PRODUCTION**. Production runs in *forward* flow with a periodic electro dialysis- reversal step (chapter 10); the Overview screen shows a countdown timer to the next reversal. TODO: the specification confirms the button sequence and mode names but does not itself state numeric pre-start interlocks/permissions (e.g. minimum tank levels) — those are defined in the PLC program, still in development.

6.4 HMI screens

Per the HMI specification, the operator interface is organized into six screens:

- **Overview** — current plant mode and Safe Stop indicator; key performance indicators (product conductivity, product flow, recovery); product routing (to customer vs. recirculation); a status badge for each stack (power on/off, current, conductivity, pressure-OK); pump run/fault status; the position of the critical reversal and CIP-loop valves; the reversal countdown timer; and the active alarm count with highest severity.
- **Tank Summary** — level and conductivity for every tank, and the state of every fill/drain valve.
- **Mode Control** — the plant-mode buttons described above, plus *Drain Concentrate Tank*, *Switch to Manual*, *ACK Alarm* and *Reset Trip*.
- **Alarm Summary** — active, acknowledged and historical alarms, each showing its code, severity, cause, effect, timestamp and a recommended action, following the ISA-18.2 alarm-management convention — see *Alarm handling* in chapter 7.
- **Maintenance / Manual** — individual device commands (pumps, valves, dosing pumps, stack power), gated by the access level in *Manual mode* below.
- **Trends** — historical KPI graphs.

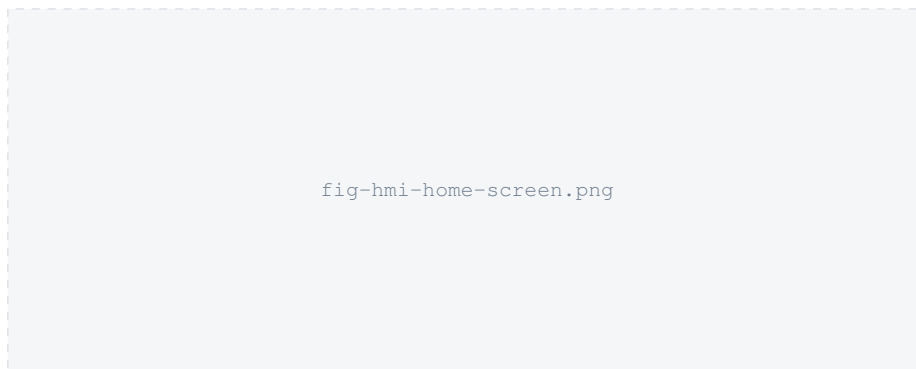


Figure 6.2. Suggested figure: screenshot of the HMI Overview screen, once the PLC program is complete and running on an actual unit, annotated with each element's function per the list above.

6.5 Reading the process values

Confirmed process readings available to the operator include pH, pressure, flow and conductivity/level for the sensor array (TSA_pH, TSA_pressure, TSA_Flow, TSA_Cond, TSA_Level), plus conductivity and level for the TB tank (TB_Cond, TB_Level) and level for the TE (electrolyte) tank (TE_Level) — see chapters 13, 15, 16 and 18. Additional electrical metering is available via power-meter tags (PM3150).

NOTE

HydroVolta's I/O tag lists follow a readable naming convention that helps when reading raw tag names on an HMI or historian trend: a leading letter indicates the measurement type (P = pressure, FL = flow), followed by c/d/e for concentrate/diluate/electrolyte, and S1–S4 for the stack number when the tag is per-stack. Tank tags use T plus a capital letter for the tank's role (TD = diluate/feed tank, TC = concentrate loop tank, TE = electrolyte tank, TB = product/breaking tank). Valve tags use VA plus a functional group number.

6.6 Changing setpoints

Per the HMI specification, every setpoint, threshold, timer, dose and recipe value is adjustable directly from the HMI at runtime, without re-downloading the PLC program. Which parameters a given user can change is governed by a three-level access hierarchy:

- **Operator:** monitor screens, acknowledge alarms, start/stop production, drain the concentrate tank, run a manual flush, view trends. Cannot change setpoints.
- **Maintenance:** everything an Operator can do, plus individual pump/valve/dosing-pump control, manual fills/drains, and non-recipe parameter changes.
- **Admin/Engineer:** everything Maintenance can do, plus stack-power control (as a set, not individually), recipe parameters, cloud command flags, and user-account management.

TODO: the specific PIN/login mechanism for each access level, and the numeric default/range for every adjustable parameter, are defined in the full HMI specification's parameter table but were not transcribed into this manual — consult HydroVolta engineering for the complete parameter table if commissioning a unit before this manual's next revision.

6.7 Manual mode

A `Manual` digital input and `LS_Select/FW_Select` mode-selection inputs exist in the control I/O map, and the HMI specification confirms **Manual** as Mode 8, reached via *Switch to Manual* on the Mode Control screen and gated to the Maintenance/Admin access levels above — an Operator-level user cannot enter Manual mode. In Manual mode, the Maintenance/Manual screen exposes direct command of every process pump, every 2-way and 3-way valve, all five dosing pumps, and stack power (as a set only, even in Manual). One routing valve (`VAR.S.5` in the specification's tag scheme) is explicitly **AUTO-only** and cannot be operated from Manual mode under any access level — do not attempt to command it manually.

6.8 Stopping the unit

A `Stop` digital input exists on the local control panel, and the HMI specification's **Controlled Shutdown** (Mode 4) is the normal, orderly way to stop production from the Mode Control screen — prefer it over simply removing power, so the system can close valves and stop pumps in a defined sequence rather than an uncontrolled one.

6.9 Emergency stop and restart

The HMI specification confirms a **hardware Emergency Stop button that bypasses the PLC entirely** — pressing it removes power/motion directly, independent of software state. TODO: its physical location on the control panel/cabinet was not confirmed in the documents reviewed; verify on the actual unit and update fig. 6.1 accordingly. Certain faults also trip the system into a software **Safe Stop** state (distinct from the hardware E-stop) — both require the following recovery sequence before production can resume, per the HMI specification:

1. Investigate the cause using the Alarm Summary screen (section 6.4) — each entry shows its cause, effect and a recommended action.
2. Physically inspect and resolve the underlying condition.
3. Press **ACK Alarm** on the Mode Control screen.
4. Press **Reset Trip** — this button is only enabled once the fault condition has actually cleared.
5. Select the next mode (e.g. Production). The PLC re-validates all permissives before allowing the mode to start, and will reject the request with a message if a required condition is still not met.



Never attempt to defeat, bridge, or otherwise bypass the hardware Emergency Stop button. If the E-stop has been pressed, do not restart the unit until the cause has been fully investigated and resolved.